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The time course of speech act recognition conveyed by speech prosody: a gating study

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ABSTRACT

Speech prosody is essential for conveying communicative intentions. Although neurophysiological data has shown that communicative functions conveyed through prosody are processed rapidly in the human brain, it is still unclear when and to what extent prosodic information is needed for the conscious speech act recognition as speech unfolds. Using a gating paradigm, we investigated the point at which listeners recognise the function of identical Italian sentences – whether they express a question or statement – based on vocal intonation. Comparing cross-spliced and natural sentences, we found that, rising or falling nuclear accentual movement on the sentence-final word seems to be the primary cue for recognition, with questions identified slightly later than statements. Furthermore, we discuss the limitations of splicing techniques in filtering out natural prosodic variations, the presence of a “statement bias” in perceiving incomplete sentences, along with a visual examination of interindividual responses. These findings offer valuable insights into the timing of conscious recognition of different communicative functions based on speech prosody.

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1. Introduction

Speech prosody is a crucial and integral feature of human verbal communication, playing a critical role in grasping the speaker's intentions and emotional state. Listeners across languages tend to rely on distinct prosodic features of speech, particularly changes in the fundamental frequency (F0) of vocal fold vibrations (Bolinger, 1978, 1986; Cruttenden, 1997), to interpret the speakers' emotional states (Banse & Scherer, 1996; Bänziger & Scherer, 2005; Frick, 1985; Mauchand & Pell, 2022; Pell et al., 2011; Simon-Thomas et al., 2009) and attitudes (e.g. Jiang & Pell, 2015; Monetta et al., 2008; Rigoulot et al., 2014). Similarly, prosodic features, including F0 changes, provide important cues for identifying the speaker's communicative function, such as suggesting, expressing wishes, or giving criticism (Hellbernd & Sammler, 2016). One well-known case where vocal intonation plays a critical role is the distinction between questions and statements. A sentence like “Mario has

three kids” can be understood as a question or a statement depending on whether the F0 contour rises or falls, typically within the sentence-final word (Bolinger, 1978; Lieberman, 1967; Srinivasan & Massaro, 2003). Note, however, that intonational marking of questions is optional in some languages, and the specific acoustic features used to differentiate questions from statements may vary cross-linguistically.

Nevertheless, across languages, the distinction between communicative functions or, in linguistic terms, so-called speech acts (Austin, 1975; Searle, 1979), such as statements and polar questions (yes/no) is often also marked by syntactic cues at various levels. In languages like English or German, this distinction is typically reflected in word order and syntactic structure, such as verb-first positioning or subject-auxiliary inversion (Ladd, 2008), though prosodic cues may also contribute. For instance, there is evidence that German speakers are sensitive to the phonetic features of

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prenuclear F0 accents when distinguishing between declarative questions and statements (see, e.g. Petrone & Niebuhr, 2014). However, in these languages, the investigation of prosodic cues is complicated by the interplay between syntactic and prosodic cues, which can be used together or separately to indicate questions. This mutual influence is, however, absent in Italian (Agard & Di Pietro, 1965; Chapallaz, 1964; Fiorelli, 1965; Maiden & Robustelli, 2014), in Spanish (Escandell-Vidal, 2012; Escandell-Vidal & Prieto, 2020), and in Portuguese (Falé & Faria, 2006; Frota, 2002), where syntactic or lexical markers are not used to signal polar questions. In these languages, prosodic features are often the primary cues for distinguishing between polar questions and statements (Maiden & Robustelli, 2014, p. 147). For instance, in Standard Italian, a sentence like *stanno lavando la macchina* – “they are washing the car” – can be understood as a statement or a polar question depending entirely on the F0 intonational contour (Agard & Di Pietro, 1965; Chapallaz, 1964; Fiorelli, 1965; Maiden & Robustelli, 2014).

The present study focuses specifically on Standard Italian, the linguistic variety that serves as a benchmark for communication and education across Italy, to investigate how intonational features contribute to the recognition of questions versus statements. The intonational features are typically described within the Autosegmental-Metrical approach (Arvaniti, 2022; Avesani, 1995; Pierrehumbert, 1981) as a sequence of low (L) and high (H) tones. These tones are organised into three key components: nuclear F0 accents, which are associated with stressed syllables; phrase accents, which define the transitional tonal movement; and boundary tones, which mark the intonational end of a phrase (Avesani, 1995). Nuclear F0 accents emphasise specific syllables, with a low F0 (L*) indicating a lower tone on the stressed syllable and a high F0 (H*) marking a rise. In Italian, accentual rises often exhibit more complex patterns, such as a low F0 minimum preceding the rise (L + H*) or a low F0 minimum at the stressed syllable followed by a high F0 peak (L* + H). Boundary tones, such as low (L%) or high (H%), indicate the edges of the phrase or the terminal intonational contour. According to this intonational description, a common intonation pattern for questions in different Italian regional varieties, from northern to southern Italy, typically includes a rising F0 accent (H*), often preceded by a low intonation (L + H*), or a low F0 accent followed by a rise (L* + H). This is then followed by a transitional low tone (L) and a final falling boundary tone (L%; full transcription: L + H* L-L%, Savino, 2012). However, some varieties also exhibit a terminal rise at the boundary tone instead of a fall (L + H* L-H%, see Table 4 in Savino, 2012). In particular, this terminal rise at the boundary tone has been often defined critical for

question recognition by a range of previous studies (Avesani, 1990; Bertinetto & Loporcaro, 2005, p. 144; and others Chapallaz, 1964; D’Imperio, 2002). Nevertheless, as highlighted by Savino (2012), the nuclear accentual rise, which is a common feature across many Italian varieties, may play a more central role in question recognition rather than the rise at the terminal boundary tone. In contrast, for statement, a typical pattern is a low nuclear F0 accent preceded by a rise (H + L*), followed by a low tone (L) and ending with a low boundary tone (L%; full transcription: H + L* L-L%, Grice et al., 2005). Thus, the encoding of intonational meaning in Italian is particularly rich in the domain of nuclear F0 accent and, in some cases also in the boundary tone, with falling and rising F0 patterns playing a critical role in question and statement recognition. While other acoustic features, such as stress, pause, speech rate and phonetic cues may also contribute (Bertinetto & Loporcaro, 2005; Cangemi & D’Imperio, 2011; Crocco, 2013; D’Imperio, 2002; Gili Fivela & Savino, 2003), this study focuses on the role of the F0 intonational contour and its role in recognising the two speech act types.

Based on these observations, Tomasello et al. (2022) measured the brain’s neurophysiological (EEG) responses in Italian native speakers to investigate when brain indexes of pragmatic prosodic markers – signalling questions vs. statements – first emerge during sentence comprehension. In this study, the sentences shared the same initial segments (e.g. “*stanno mangiando*”, “*they are eating*”), with only the final noun (e.g. “*la pasta*”, “*the pasta*”) differing in F0 contour: one with a rising-falling F0 at the nuclear accent followed by a second rise at the boundary tone (L + H* L-H%) and the other with a falling nuclear F0 followed by a low boundary tone (H + L* L + L%), in line with previous observations. To ensure the intonational difference was confined/isolated to the final word, while controlling for other prosodic and acoustic features of the sentence, 50% of the articles “*la*” preceding the final noun in a given sentence type were swapped and spliced into the respective other sentence type, and vice versa. The results reveal that distinct brain responses emerged as early as 100 ms after the onset of the sentence-final word in Italian sentences specifically aligned around the nuclear accentual rise or fall movements (see Fig. 3A in Tomasello et al., 2022). This suggests that communicative functions conveyed by F0 prosodic contour differences at the sentence final word are processed rapidly in the brain. Crucially, these brain response differences were not merely driven by acoustic variations, as multiple control conditions were implemented to account for neural responses to physical acoustic differences in the speech signal (Tomasello et al., 2022).

The rapid processing observed in the human brain aligns with a range of previous neurocognitive studies demonstrating the early emergence of neurophysiological responses during the processing of communicative functions in different modalities (spoken, written, and gestural context; see Antoine et al., 2024; Boux et al. 2021; Egorova et al., 2013, 2014; Tomasello et al., 2019; for a review, see Tomasello, 2023). Other studies have examined the cortical locus of prosodic intonation processing. For example, a study by Sammler et al. (2015) examined rising and falling vocal tones signalling questions and statements in the context of single words. Their findings revealed a dual route system in the right hemisphere, involving both the ventral and dorsal streams, auditory processing along the superior temporal lobe and within motor regions. Furthermore, another study demonstrated the functional interaction between auditory and social brain regions, areas of the theory-of-mind (ToM) network relevant to understanding the mental states of others, was shown to be engaged in decoding the perceived prosodic tones into speaker intentions (Hellbernd & Sammler, 2018; for a review see Tomasello et al., 2025).

Even though previous behavioural and neurocognitive studies have shed light on the role of prosody in decoding a speaker's communicative intention and identified related brain correlates, they cannot fully clarify how much information is needed to accurately recognise different speech acts, as speech unfolds. Although the instantaneous neural processing of questions and statements conveyed by speech prosody, as assessed using the EEG (Tomasello et al., 2022) offers insight into the early stages of understanding processing, it may not directly align to the exact moment at which listeners “consciously” recognise the speaker's communicative intention. This recognition is thought to occur within the final critical word; however, whether at the nuclear accentual rise (L+H*), as pointed out by Savino, 2012 or the rise at the boundary tone (L-H%), traditionally discussed as being critical for question recognition (e.g. Avesani, 1995; D'Imperio, 2002) remains an open question. Here, the term “recognition” refers to the ability to *consciously* distinguish between speech acts conveyed by prosody (see e.g. for similar research Holtgraves, 2008; 2021).

Moreover, in most previous neurocognitive studies examining speech prosody for linguistic information processing at various levels (phonology, semantics, syntax, etc.), the acoustic stimuli are commonly spliced, combining different recordings of the same sentence or sound sequence to ensure minimal phonological differences between the stimuli. This technique is used

to control for variation in acoustic properties within specific speech segments (Astésano et al., 2004; Kotz & Paulmann, 2007; Steinhauer et al., 1999; Tomasello et al., 2022). Although splicing is a widely used technique, it can also compromise the naturalness of the stimuli and often disrupts the continuity of phonological intonation patterns (for a discussion, see Steinberg et al., 2012). Critically, a recent study analysing the Corpora and Lexicons of Spoken and Written Italian (CLIPS) has pointed out that the discrimination between statements and questions may be influenced by the F0 contour of the entire sentence, and not just the nuclear accentual F0 contour of the critical final word of the sentence (D'Andrea et al., 2014). Similar results have also been reported in English (Lieberman, 1967; O'Shaughnessy, 1979; Studdert-Kennedy & Hadding, 1973), suggesting that significant differences in F0 contour relevant to recognising the two speech acts may emerge earlier in the sentence, before the terminal part of an utterance. Thus, yet another critical question arises: Does the recognition point of a speech act during the temporal processing of a spoken sentence differ between well-matched spliced stimuli and naturally spoken sentences? Crucially, new efforts are needed to investigate the time course of recognition of communicative functions based on F0 contour and to compare these findings with previous neurocognitive data, stimuli, and results (Tomasello et al., 2022).

An effective approach to investigating the temporal course of speech sounds and thus examining the recognition point during the unfolding of speech information is the gating paradigm (Grosjean, 1980). Traditionally, the gating paradigm has been used in spoken word recognition to study lexical access and to investigate the amount of acoustic-phonetic information required to identify a stimulus (Grosjean, 1985; Marslen-Wilson, 1987; Marslen-Wilson & Welsh, 1978; Salasoo & Pisoni, 1985). In the original version of the gating task (Grosjean, 1980; Howes, 1957; Marslen-Wilson & Welsh, 1978; Rosenzweig & Postman, 1957; Rubenstein & Pollack, 1963; for a review see Grosjean, 1996; Marslen-Wilson, 1984), participants listen to a series of auditory “gates”, where each gate presents incrementally longer portions of a spoken element (e.g. syllable, word, or sentence). The increment size of the gates can be expressed either as a percentage of the target element or in milliseconds. At each presentation, the length of the acoustic segment is progressively increased by a constant amount (for instance, first 200 ms, then 400 ms, and so on), starting from the beginning of the relevant stimulus. The final gate usually corresponds to the complete stimulus. After each incremental presentation, listeners are asked to

identify the target element and to report their confidence level of their response. This technique allows researchers to estimate the point at which a stimulus is accurately recognised by a participant without any subsequent changes in response, as well as the point at which the participant feels most confident of having correctly identified the stimulus (“recognition point”, Grosjean, 1980). The gating task has been used, for instance, to examine word frequency effect (Grosjean, 1980; Metsala, 1997; Walley et al., 1995) or to explore emotional meaning conveyed by music (Vieillard et al., 2008) as well as emotional expression in speech (see Audibert et al., 2007; Pell & Kotz, 2011). For instance, Pell and Kotz (2011) found that some emotional prosodies were recognised in earlier gates than others, relative to the onset of the utterances (gate 1): fear was identified around 500 ms, anger at 700 ms, and happiness at 1000 ms. Recently, it has been shown that such a paradigm also works for non-verbal emotional vocalisation (Schaerlaeken & Grandjean, 2018), further demonstrating its broad applicability. Given these successful adaptations, the gating paradigm seems to be a promising technique for examining the time course of speech act recognition based on prosody, a topic that remains relatively unexplored to date.

In the present study, we explored the role of the F0 contour in the recognition of question and statement functions in spoken Standard Italian, by using a subset of the stimuli from Tomasello et al. (2022). The aim of the present study was twofold. First, to document the recognition of speech acts at different time points (measured in so-called gates) as the temporal information of speech unfolds, and thus to determine how much prosodic information is needed for the accurate classification of questions and statements. Second, to explore whether any (dis)similarities would emerge between recognition points for questions and statements in cross-spliced stimuli, versus unaltered natural spoken sentences. Note that a direct comparison between cross-spliced and natural stimuli is challenging, as they differ in several ways (see sec. 2.4 below), thus only an indirect comparison within each condition was feasible. Furthermore, we employed short gates of fixed 50 ms to obtain accurate temporal information on recognition points and to ensure compatibility with the constant temporal resolution typical of EEG recordings, facilitating a more precise comparison with findings from a prior EEG study (Tomasello et al., 2022). This contrasts with other studies with similar research aims, which used larger gating increments of 100 ms (Audibert et al., 2007), 250 ms (e.g. Cornew et al., 2010) or linguistic unit of around 300 ms (Pell & Kotz, 2011).

We hypothesised that the accuracy of speech act recognition would increase systematically as more of the acoustic signal is revealed at successive gate intervals. Specifically, building on previous studies described above (Savino, 2012), we assume that listeners would be able to reliably distinguish between statements and questions at the final segment for the cross-spliced sentences, specifically at the nuclear accentual rise or fall, rather than at the rise in the boundary tone. Assumptions about earlier differences before the final segment of the sentence cannot be made in the cross-spliced sentences, as most gates are located on the final word. However, differences are unlikely to emerge, as both speech acts share the same initial segment. For natural sentences, we expect clear recognition to occur at the nuclear accentual modulation. However, recognition differences may occur earlier due to pre-final F0 variations, which we further investigated through finer gate divisions.

2. Methods

2.1. Data availability

Data and scripts are made publicly available in the following Open Science Framework repository: <https://osf.io/c83ep/>

2.2. Participants

Sixteen healthy volunteers (mean age 27 years; range: 19–35; 9 female) took part in the study. All participants were Italian native speakers, from different Italian regions (i.e. #1 Tuscany, #2 Friuli, #3 Emilia-Romagna, #4 Lazio, #5 Lombardy, #6 Campania, #7 Apulia, #8 Lazio, #9 Sardinia, #10 Trentino Alto Adige, #11 Calabria, #12 Liguria, #13 Marche, #14 Abruzzo, #15 Tuscany, #16 Emilia-Romagna, in total this included 4 from northern, 8 central, and 4 southern Italian regions), and had no history of hearing difficulties, as well as neurological or psychiatric diseases. Procedures were approved by the Ethics Committee of the Charité Universitätsmedizin, Campus Benjamin Franklin (Berlin, Germany) (Protocol n. EA4/138/19 approved on 03 September 2019). All participants were informed that the study investigated the perception of Italian sentences, but they were naïve as to the specific predictions of the experiment. They were paid € 50 for participating in the whole experiment that lasted approximately 2–3 h and signed an informed consent form prior to the start of the experiment.

2.3. Stimulus recording and selection

Cross-spliced stimuli: A subset of fifteen spoken sentences were selected from Tomasello et al. (2022) to be

used as experimental stimuli. Stimuli were recorded (44,100 Hz, 32-bit float, mono) in Standard Italian by a female adult Italian native speaker, originally from Sicily (i.e. southern Italy). These sentences were all comparable in their underlying syntactic structure as they consisted of a third-person plural verb in *perifrasi passiva* construction (similar to the English present continuous) then followed by a singular, feminine, determinate article and by a corresponding singular feminine noun (e.g. “*stanno prendendo la metro*”, “*they are taking the metro*”). The subset of sentences used in this study included 12 disyllabic (e.g. *me-tro*) and 3 trisyllabic final words (e.g. *macchi-na*) denoting familiar everyday objects (see Appendix 1). For the recordings, the first two verbs (e.g. “*stanno* and “*prendendo*”, “*they are* and “*taking*”) were recorded separately with a neutral intonation. By neutral, it is meant that the speaker was instructed to pronounce the initial segments of the sentence avoiding any specific intentions, emotional inflections, or emphasis. The first and second verb in the initial segment resulted in an H + L* L + L% declarative intonational pattern (i.e. falling contour), despite the speaker was explicitly directed to use neutral intonation. This might be due to the isolated recording. The article and the critical noun (e.g. “*la metro*”, “*the metro*”) were recorded together, each produced as both a statement and a question. Here, we instructed the speaker to imagine being engaged in a conversation with a friend and to utter all critical final words sequentially with the preceding article as if requesting information (question), and then as if informing the partner about facts or events (statement). The resulted intonational contour revealed distinct patterns of rising and falling intonational patterns. The rising intonation pattern was characterised by a L + H* L-H% contour (i.e. raising-falling contour), where a low tone was followed by a rising F0 on the stressed syllable, a low transitional tone, and a high boundary tone, typically associated with interrogative intent. In contrast, the falling intonation pattern followed a H + L* L + L% contour (i.e. falling contour), with a high tone transitioning to a low F0 on the stressed syllable, followed by a falling F0 and a low boundary tone, signalling declarative meaning. The article (“*la*”, “*the*”) was then cross spliced between the two speech acts, with 50% of the articles swapped between questions and statements. Although the use of 15 sentences introduced a minor imbalance in the cross-splicing (resulting in one article not evenly distributed), this slight discrepancy is not expected to significantly affect the findings, as an additional analysis by excluding trisyllabic words showed similar results (see more below). These elements were combined to create the stimuli, with the first two words (“*stanno* and “*prendendo*”, “*they are* and “*taking*”), the cross-spliced article

(“*la*”, “*the*”), and the final noun (“*metro*”) characterised by a rising and falling F0 contour, as described above. This was done to make sure that the main differences in prosody were confined to the sentence-final word, while the sentence segments preceding it were matched for acoustic features. The sentence length was on average 3397 ms (min. 3300 ms, max. 3450 ms), and a silent pause was included between 700 and 1000 ms prior to the article onset. The pause was added specifically for the EEG study to increase the signal-to-noise ratio and specifically to separate the brain responses of the preceding segment to that of the critical final word, which is not relevant for the present study (for more details, see Tomasello et al., 2022). We decided to reuse the original set of stimuli, for a more direct comparison between datasets and to further investigate whether a longer pause before the critical point might affect recognition. We refer to this set of stimuli as *cross-spliced sentences* (see Figure 1).

Natural stimuli: To examine whether the splicing technique leads to interference effects or differences in recognition points between the two speech acts, the same fifteen spoken sentences were recorded in another recording session by the same speaker and uttered as naturally as possible at a natural pace, once as a question the second time as a statement. The final critical words exhibited distinct intonational patterns for questions and statements. In questions, the stressed syllable featured a low F0 followed by an immediate rise, with a low transitional and boundary tone (L*+H L-L%, rising-falling contour). In statements, a high tone transitioned to a low F0 on the stressed syllable, followed by a low F0 and a low boundary tone (H + L* L + L%, i.e. falling contour). The pre-final F0 contour revealed a notable difference, with a higher F0 peak at the second stressed syllable for questions compared to statements at the second verb (“*stanno prendendo la metro*” – “*they are taking the metro*”), while both speech acts exhibited the same L*+H L-L% pattern. The length of the natural sentence was on average 1631 ms (min. 1520 ms, max. 1720 ms), and acoustic parameters were not manipulated after recording, except for minimal changes done using PRAAT 6.0.49 speech analysis software (<http://www.praat.org>, Boersma, 2001). Specifically, only a small portion of silence between the verb and the article was removed in order to align the article’s onset within all natural sentence stimuli. This adjustment was made to ensure consistency in the onset of the last sentence segment and gate construction between two sentence types (see Figure 1). Finally, all the sentences were normalised to the same average sound energy and cleaned from background noise using Audacity 2.2.2. software (for the

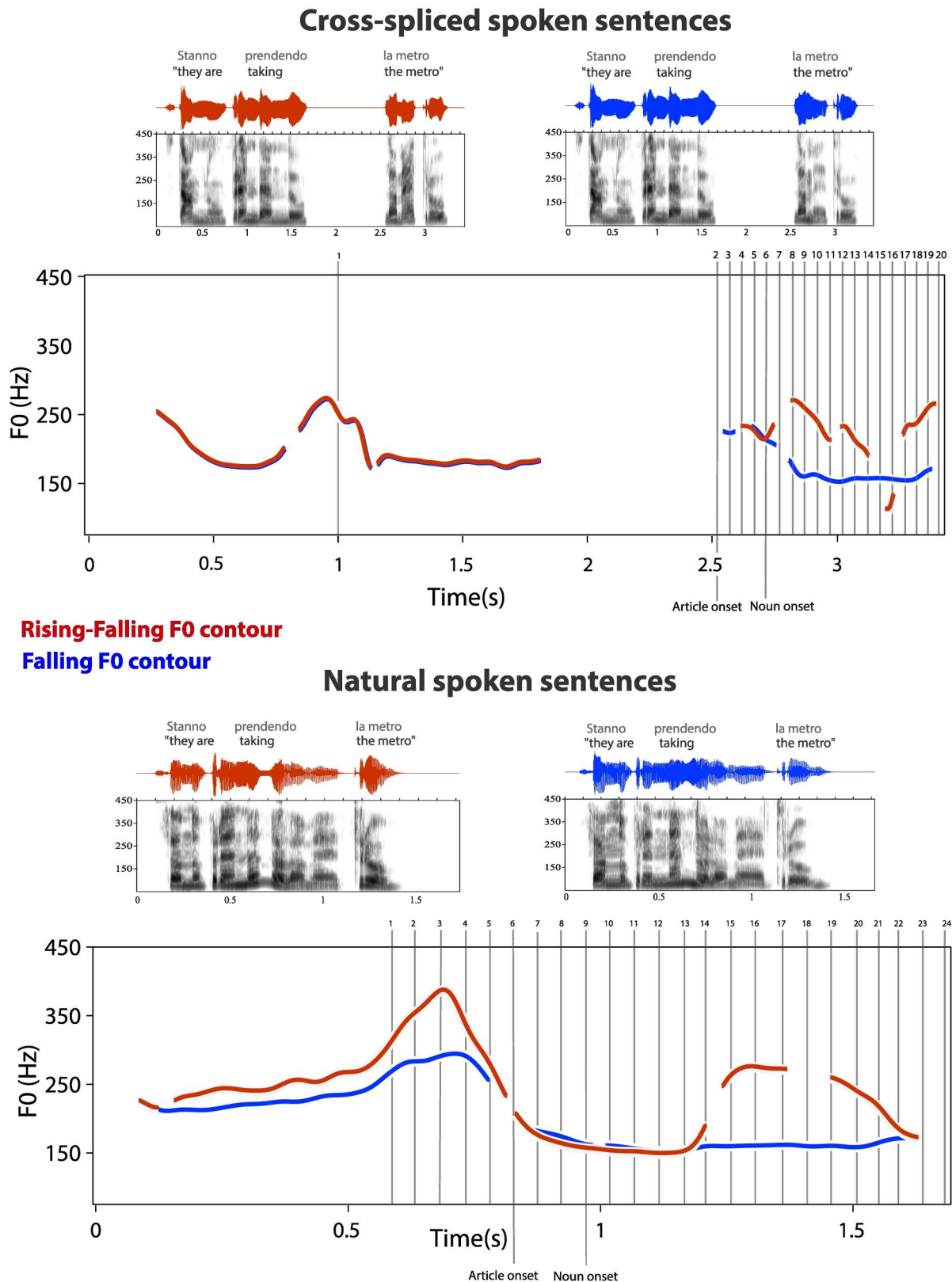


Figure 1. The figure shows speech waveform and spectrogram plotted for one stimuli example (upper panel: left for question, right for statement), along with the average F0 contour (lower panels). Statement (in blue) exhibit a falling contour ($H + L^* L + L\%$) on the final noun in both cross-spliced and natural stimuli, corresponding to the lower line in the figure. Question (in red) shows a rising-falling contour on the final noun, represented by the higher line, with an additional rising at the boundary tone ($L + H^* L-H\%$, rising line) in cross-spliced stimuli and a rising-falling contour with low boundary tone ($L^*+H L-L\%$) in natural stimuli (for more details see Sections 2.3 and 2.4). The vertical grey lines indicate the maximal duration of the different gates probed. Note that, in addition to the intonational difference after the splicing point of the well-controlled sentences at/around the final noun onset, the naturally spoken sentences show intonational differences starting at, or even before, gate 1 (within the main verb).

noise removal algorithm, see source code at <https://github.com/audacity/audacity>).

2.4. Differences and similarities between cross-spliced and natural spoken sentences

Duration & pause: The two conditions differed in duration, with cross-spliced sentences lasting on average 3397 ms and natural sentences lasting on average 1631 ms. This difference is due to the splicing technique and particularly to the longer pause prior to the critical noun onset in cross-spliced sentences that as mentioned above are often necessary in neurocognitive experiments. Although this discrepancy renders direct comparisons between the two conditions difficult, the study specifically examines how splicing, and the extended pause affect sentence recognition. In particular, it investigates how rising and falling F0 contours at the nuclear accentual tone of the final word influence the perception of questions and statements within each condition (cross-spliced and natural). Importantly, the same recorded spoken sentences with identical speech content and syntactical structure were used across both conditions and for both speech acts.

Prefinal F0 contour: Within each condition, statements and questions differed mainly in the F0 contour at the prefinal F0 contour. In natural sentences, questions exhibited a more pronounced F0 rise at the second stressed syllable compared to statements at the second verb (e.g. “*stanno **pren-DEN-do** la metro*”, “*they are **taking** the metro*”). This distinction was absent in the cross-spliced condition, where the first sentence segment was identical for both speech acts.

F0 final word contour: Questions, in both conditions and on average across all stimuli, show a rising-falling F0 contour at the nuclear accents, along with differences at the boundary tone. The cross-spliced condition exhibits two distinct F0 rises: one at the accentual stressed syllable and another at the boundary tone (L + H* L-H%). In contrast, the natural condition features a single F0 rise localised directly after the accentual syllable and a low boundary tone (L*+H L-L%, see Figure 1). However, note that both conditions share common intonational patterns for questions at the nuclear F0 accent or the immediately following segment aligning with those reported in previous studies (Savino, 2012). These differences may stem from the recording process: sentence-final words were recorded in isolation in the cross-spliced condition and within the full sentence context in the natural condition. Statements in both conditions consistently followed an intonation contour (H + L* L + L%, i.e. falling contour) that is characteristic of a typical declarative intonational pattern.

Intensity: Across both the cross-spliced and natural conditions, another key difference between the two speech acts lies in the greater intensity of the article “la” (the) preceding the final target word in statements compared to questions. This was confirmed by the Wilcoxon matched-pairs signed-rank test, which showed a significant difference between the two speech acts (natural condition, $Z = 3.06$, $p = .0021$, cross-spliced, $Z = 3.35$, $p = .0008$, 150 ms from article onset).

2.5. Gate construction

Each sentence was edited using Audacity 2.2.2. software to produce acoustic segments or “gates” that varied as a function of increasing duration. Since cross-spliced and natural sentences differed in average length (cross-splicing 3397 ms; natural 1631 ms), we defined our gate incrementation according to the duration of each sentence and the article and word onsets. Importantly, the latter were aligned, starting at the same time point across all sentences within each condition.

For the cross-spliced stimuli, the first gate consists a 1000 ms segment corresponding to the first verb and the first syllable of the second verb (e.g. “*stanno pren*”, “*they are ta*”), and a second gate extended to 2550 ms, corresponding to the article onset “la” in all sentences. Subsequently, the length of the acoustic segment was increased by a constant of 50 ms until the end of the stimulus duration. The reason for shorter gates only from the article onset lies on the design of the cross-spliced stimuli, where the first part of the sentence segment was exactly the same between the two speech act types, and thus no differences would be expected. Consequently, potential differences were expected to emerge from the article onset onwards and specifically whether at the nuclear accentual rise or later at the boundary tone in the final word of the sentence. Adopting a fixed gating paradigm ensures compatibility with the constant temporal resolution typical of EEG recordings, facilitating a more precise comparison with findings from a prior EEG study (Tomasello et al., 2022). The two initial long gates (gates 1 & 2) covering the first sentence segment (e.g. “*stanno prendendo*”, “*they are taking*”) had the function of sanity checks to make sure that participants do not notice any differences prior to the sentence-final word. We obtained a range of 17–20 gates for sentences in cross-spliced conditions (min. 3300 ms, max 3450 ms). The range of the gates varied between sentences, as some of them were longer or shorter, so the gates were divided individually for each sentence. The longest cross-spliced stimulus was 3450 ms long, shorter stimuli were padded with silence to the same length, and then the

shorter stimuli had their longest gates not presented. However, the article and noun onset occurred at the same time in all sentences, as they were meticulously aligned during sentence creation. Gate 2 corresponds to the article onset, gate 6 to the final noun onset, and approximately at gate 8/9 on average to the nuclear accentual syllable (see Figure 1).

For the natural sentences, the gate division was adapted to account for their shorter duration and to ensure the inclusion of the F0 contour differences observed between the questions and statements in the second verb segment (e.g. “*stanno prendendo*”, “*they are taking*”), as well as in the sentence-final word (see Section 2.4 and Figure 1). To this end, the first gate was until 570 ms, corresponding to the second main verb onset and from here gates increased by a constant of 50 ms until the end of stimulus duration. For this condition, we obtained a range of 20–24 gates for each sentence (min. 1520 ms, max. 1720 ms). Gate 2 corresponds to the second main verb onset, gate 6 to the article onset, gate 9 to the noun onset, as all sentences word were aligned after recordings. On average, gate 16/17 correspond approximately to the nuclear accentual rise/fall at the stressed syllable. Note that gate divisions from the article onset “la” were exactly the same for both cross-spliced (from gate 2) and natural (from gate 6) stimuli, thus making both comparable at the sentence-final word. Table 1 presents the mean F0 values for each gate across the two speech acts in each condition, allowing comparison with the average responses from the discrimination tasks. However, it is important to note that the averaged mean F0 values shown in the table may not precisely align with stress patterns, number of syllables, and their position, as these features vary across the stimuli. However, since article and word onsets are consistently aligned, and 13 of the 15 words were disyllabic, the alignment of the stressed syllable can be approximated within the table, with variability of up to 1 gate for disyllabic words and up to 2 gates for trisyllabic words. Thus, for the purposes of this experiment, the aggregated data would provide sufficient insights to address the research questions and examine the overall patterns of speech act recognition.

2.6. Experimental procedure

Each participant was tested individually in a quiet room at the Brain Language Laboratory Freie Universität Berlin. The experiment was set up and run in E-prime 2.0 (Psychology Software Tools, Pittsburgh, PA) and stimuli were presented through headphones at a comfortable intensity level. The experiment consisted of two blocks, whose order was counterbalanced across

subjects. One block included either cross-spliced or natural sentences of 15 Italian spoken sentences with both rising and falling F0 contour at the final word sentence (in total 30 sentences) and was presented in a unique pseudorandom order across participants. All gate segments of each sentence (ranging from a minimum of 17 to a maximum of 24) were presented incrementally, one after the other, before moving on to the next sentence. Each sentence always began with gate 1 and then proceeded as a function of specific segments of increasing duration, as described above, concluding with the last gate that corresponded to the entire stimulus event (see the previous section).

During the experiment, each segment of the same sentence was played a single time, and participants were instructed to make two judgments in sequence.

Table 1. Mean of fundamental frequency (F0) for each gate interval is shown in table for both speech act types (i.e. questions: rising-falling contour, statements: falling contour) in cross-spliced and natural conditions. “AO” article onset; “NO” noun onset, which are consistently aligned across all stimuli within each condition. Major F0 changes are highlighted, and on average, correspond to the nuclear accentual rise/fall or immediately after. This alignment is approximate, as the stimuli varied within each condition. However, the variations were minimal, as 13 out of 15 stimuli were disyllabic, allowing for reasonable approximation, with variability of up to 1 gate for disyllabic words and up to 2 gates for trisyllabic words. Gate 24 in the natural stimuli does not exhibit a measurable F0, as the final segment contains only minimal vocalisation along with a pause. For more details, please refer to the audio files and gates in the repository link below.

Cross-spliced				Natural			
gates	ms	falling mean F0(Hz)	rising- falling mean F0 (Hz)	gates	ms	falling mean F0(Hz)	rising- falling mean F0 (Hz)
1	1000	212.2	211.3	1	570	217.9	244.8
–	–	–	–	2	620	253.3	318.3
–	–	–	–	3	670	275.1	355.4
–	–	–	–	4	720	282.5	398.8
–	–	–	–	5	770	292.7	318.1
2 AO	2550	191.6	192.8	6 AO	820	266.5	267.1
3	2600	223.4	204.1	7	870	235.6	194.9
4	2650	209.5	220.1	8	920	178.3	169.3
5	2700	232.0	231.1	9 NO	970	172.0	158.8
6 NO	2750	214.0	210.0	10	1020	157.8	155.0
7	2800	213.2	237.9	11	1070	154.8	149.9
8	2850	177.3	273.1	12	1120	152.2	149.8
9	2900	160.7	262.3	13	1170	148.0	149.4
10	2950	164.9	247.8	14	1220	148.8	157.4
11	3000	156.2	207.8	15	1270	157.7	244.4
12	3050	153.4	222.3	16	1320	155.6	278.5
13	3100	159.0	217.5	17	1370	157.1	279.1
14	3150	158.4	206.2	18	1420	158.0	306.4
15	3200	159.9	209.6	19	1470	155.0	256.7
16	3250	158.3	187.4	20	1520	157.9	252.2
17	3300	155.5	222.7	21	1570	152.3	216.5
18	3350	157.1	228.6	22	1620	164.9	179.9
19	3400	172.4	259.9	23	1670	168.0	169.9
20	3450	174.8	266.7	24	1720	0	0

First, participants had to provide 3-alternative forced choices about the communicative function of each gate. They were asked to respond to the question “Cosa vuole comunicare la persona che ha prodotto questo suono (sequenza)?” (“What does the person who uttered this sound (sequence) want to communicate?”) by choosing with the mouse one of the three possible options presented on the screen: “affermazione”, “domanda”, “non è chiaro” (“statement”, “question” or “unclear”). Upon responding, they had to rate how certain they were of their response on a continuous scale from 0 (unsure) to 100 (sure), presented vertically with 0 on the bottom and 100 on the top. At the end of each block, to examine how natural the cross-spliced and natural acoustic stimuli sounded participants listened to the entire set of sentences (question and statement) once again, which were presented in randomised order. For each sentence, they had to rate how natural it sounded. They had to indicate their answer on a continuous vertical scale from 0 (unnatural) to 100 (natural). Participants were informed that naturalness refers to how natural the sentence sounded if heard in real communicative interaction.

3. Analyses

Data analysis and visualisation were carried out with R (R Core Team, 2019) and RStudio (v4.2.2.). To examine differences in the time course of speech act recognition between questions and statements within each condition, we measured the frequency of responses selected by participants (i.e. question, statement, unclear) at each gate interval for each speech act (i.e. questions, statements) in cross-spliced and natural conditions and converted these frequencies to percentages. These results were combined with the confidence score, which involved participants indicating how certain they were about their decision at each gate, and 70% was taken as the confidence threshold value. We chose 70%, as 60% and 80% of confidence scores showed similar results (see below), as well as to ensure that participants’ confidence was neither too close to chance (60%) nor too exact (80%). The combination of response frequency and confidence scores was used to determine the recognition point with greater accuracy, as it is assumed that incorporating confidence scores of a given answer or judgement may reveal the actual recognition point during speech understanding (Cornew et al., 2010). Specifically, during data preprocessing, we classified the answer given for each gate as a question or statement, when their confidence response was above $\geq 70\%$, while the responses that had confidence

scores below $< 70\%$ were all re-classified as “unclear” along with option response unclear. The confidence ratings collected for the unclear option were not taken into account, as this value does not provide any additional information regarding their decision.

The described frequency of classification was analysed in the cross-spliced and natural condition separately via repeated-measures analyses of variance (ANOVAs) with the factors *Response* (three levels: questions, statements and unclear), *F0 contour* (two levels: rising-falling and falling), and *Gate* (20 levels for cross-spliced stimuli; 24 levels for natural stimuli). Since stimulus durations varied (17–20 gates for cross-spliced and 20–24 for natural stimuli), the final gate levels included data from fewer time points (4–6, depending on the condition), whereas earlier gates incorporate responses from up to 15 items. Nonetheless, all subjects contributed data to all gates, as the ANOVA was conducted on the aggregated dataset. However, note also that recognition is expected to occur earlier, within gates that include aggregated responses from all stimuli, rather than relying solely on the final gates, where fewer data points are available.

To further investigate any significant interactions, we ran post-hoc tests with Tukey’s HSD correction (Tukey, 1977) for multiple comparisons to identify the significant differences in the frequency of responses in each gate interval of questions and statements. To this scope, the functions *anova_test()* and *tukey_hsd()* from the R package *rstatix* were used. For conciseness, we report in text only significant differences in pairwise comparisons at the point at which participants were confident of having recognised the speech act. We will consider the Recognition Point (RP, see Grosjean, 1980; 1996; Walley et al., 1995) as the point (measured in gate number or milliseconds from sentence onset) in which a stimulus classification as a statement or question prevails over the other two options, as indicated by statistically significant differences. Moreover, an additional statistical analysis was conducted considering only disyllabic sentence-final nouns with the stress in the first syllable, which comprised 12 out of 15 items, and removed the trisyllabic words – two with stress in the second syllable or one on the first. This additional analysis was conducted to ensure that the overall results remain consistent and are not influenced by the inclusion of trisyllabic words. It also serves to validate the robustness of the findings, indirectly addressing the slight imbalance in the cross-spliced article within the cross-spliced conditions.

Finally, the naturalness scores data were submitted to a 2×2 repeated-measures analysis of variance (ANOVA) with the factors *F0 contour* (two levels: rising-falling and

falling) and *Condition* (two levels: cross-spliced and natural). The Greenhouse Geisser correction was applied to all effects involving factors with more than 2 levels. The corrected *P*-values and epsilon (ϵ) were calculated as implemented in STATISTICA 12 (Statsoft, Tulsa, OK, USA) and reported throughout the paper where applicable. Partial eta-square (ηp^2) values are also stated, which is defined as an index of effect size (0.01–0.06 small, 0.06–0.14 medium, and >0.14 large; Cohen, 1988). The percentages reported in the Results section refer to the frequency of responses selected by participants for the three response options (i.e. statement, question, unclear), after re-classifying the data based on confidence scores described above.

4. Results

Cross-spliced condition: The results of the repeated-measures ANOVAs show a significant interaction between *Response*, *F0 contour*, and *Gate* ($F(38, 570) = 9.84$, $p \leq 0.001$, $\eta p^2 = 0.87$, $\epsilon = 0.149$). Post hoc Tukey's tests indicate that for sentences with a rising-falling F0 contour with an extra rise at the boundary tone within the final word (L + H* L-H%), the responses as questions become significantly more frequent from Gate 12 (3050 ms) (Mean, $M = 58\%$, Standard Error, $SE = 0.05$) than statement ($M = 12\%$, $SE = 0.02$; $p < 0.001$) and unclear responses ($M = 31\%$, $SE = 0.05$; $p < 0.001$). While for sentences with a falling F0 contour within the final word (H + L* L-L%) the responses as statements become significantly more frequent from Gate 9 (2900 ms) ($M = 63\%$, $SE = 0.07$) than questions ($M = 4\%$, $SE = 0.01$; $p < 0.001$) and unclear responses ($M = 33\%$, $SE = 0.06$; $p = 0.002$) (Figure 2). When comparing only the frequency responses of questions and statements, in the rising-falling F0 contour (L + H* L-H%) we observed the following pattern: the responses as statement were more frequent from Gate 1 to Gate 7 ($M = 34\%$, $SE = 0.07$; $p = 0.054$), while the responses as question were more frequent from Gate 10 until the end ($M = 39\%$, $SE = 0.06$; $p = 0.0053$). For the falling F0 contour (H + L* L-L%), the responses as statement were more frequent from Gate 1 ($M = 32\%$, $SE = 0.09$; $p = 0.010$) onward (see Supplementary materials Excel file for all post-hoc pairwise comparisons).

Natural condition: Similarly, the results of the repeated-measures ANOVAs for the natural stimuli also show a significant interaction between *Response*, *F0 contour*, and *Gate* ($F(46, 690) = 26.088$, $p < 0.001$, $\eta p^2 = 0.64$, $\epsilon = 0.107$). Post hoc Tukey's tests showed that for sentences with a rising-falling F0 contour within the final word (L* +H L-L%), the responses as questions become significantly more frequent from Gate 17 (1370 ms) ($M = 51\%$, $SE = 0.07$) than statements ($M = 21\%$, $SE = 0.06$; $p = 0.004$) and unclear ($M = 28\%$, $SE = 0.06$; $p = 0.032$). While

for sentences with a falling F0 contour within the final word (H + L* L-L%), the responses as statements become significantly more frequent from Gate 13 (1170 ms) ($M = 64\%$, $SE = 0.08$) than questions ($M = 3\%$, $SE = 0.01$; $p < 0.001$) and unclear ($M = 33\%$, $SE = 0.08$; $p = 0.003$) (Figure 3). When comparing only the frequency of responses for questions and statements, in the rising-falling F0 condition (L*+H L-L%) we observed the following pattern: the responses as statement were slightly more frequent at Gates 1 ($M = 19\%$, $SE = 0.07$; $p = 0.08$), while questions were more frequent from Gate 5 until Gate 11 ($M = 31\%$, $SE = 0.08$; $p = 0.032$), and again from Gate 16 until the end ($M = 31\%$, $SE = 0.08$, $p = 0.032$). For the falling F0 contour (H + L* L-L%), the responses as statements were more frequent from Gate 1 ($M = 29\%$, $SE = 0.09$; $p = 0.01$) onwards, similarly as for the cross-spliced condition (see Supplementary materials Excel file for all post-hoc pairwise comparisons).

The analyses considering 60% and 80% thresholds yielded converging results with our main findings at 70% threshold, with only minor differences (i.e. variations of 1 or 2 gates in recognition point), however, all are still within the critical final words. Likewise, the results of the analysis excluding trisyllabic words revealed similar recognition points in both conditions. The full report can be found in the OSF repository (please see Appendices 2A, 2B, and 3, and the Supplementary Materials Excel file for post-hoc pairwise comparisons).

Naturalness results: The results of naturalness ratings showed significant main effects of *Condition* ($F(1, 15) = 12.281$, $p = .003$, $\eta p^2 = .217$) and *F0 contour* ($F(1, 15) = 22.869$, $p < .001$, $\eta p^2 = .145$). Overall, stimuli in cross-spliced condition obtained lower naturalness scores ($M = 57.23$, $SE = 5.56$) compared to those in natural condition ($M = 77.99$, $SE = 5.26$) and sentences with a falling F0 contour (i.e. statements) were rated as more natural than sentences with a rising-falling F0 contour (i.e. questions) ($M = 75.73$, $SE = 5.23$; $M = 59.49$; $SE = 6.05$, respectively). However, we found no significant interaction between F0 contour and Condition ($F(1, 15) = .176$, $p = .681$, $\eta p^2 = .001$). This suggests that questions were perceived as less natural than statements regardless of cross-spliced (questions $M = 49.79$, $SE = 5.51$; statement $M = 65.68$, $SE = 5.12$) or natural (questions $M = 69.20$, $SE = 5.73$; statement $M = 86.78$, $SE = 3.75$) sentences. Descriptive statistics and plots of naturalness ratings split by subjects' regional varieties can be found in Appendix 5.

5. Discussion

The present study aimed at examining the temporal dynamics of speech act recognition of questions and statements conveyed by speech prosody. Specifically,

Cross-spliced spoken sentences

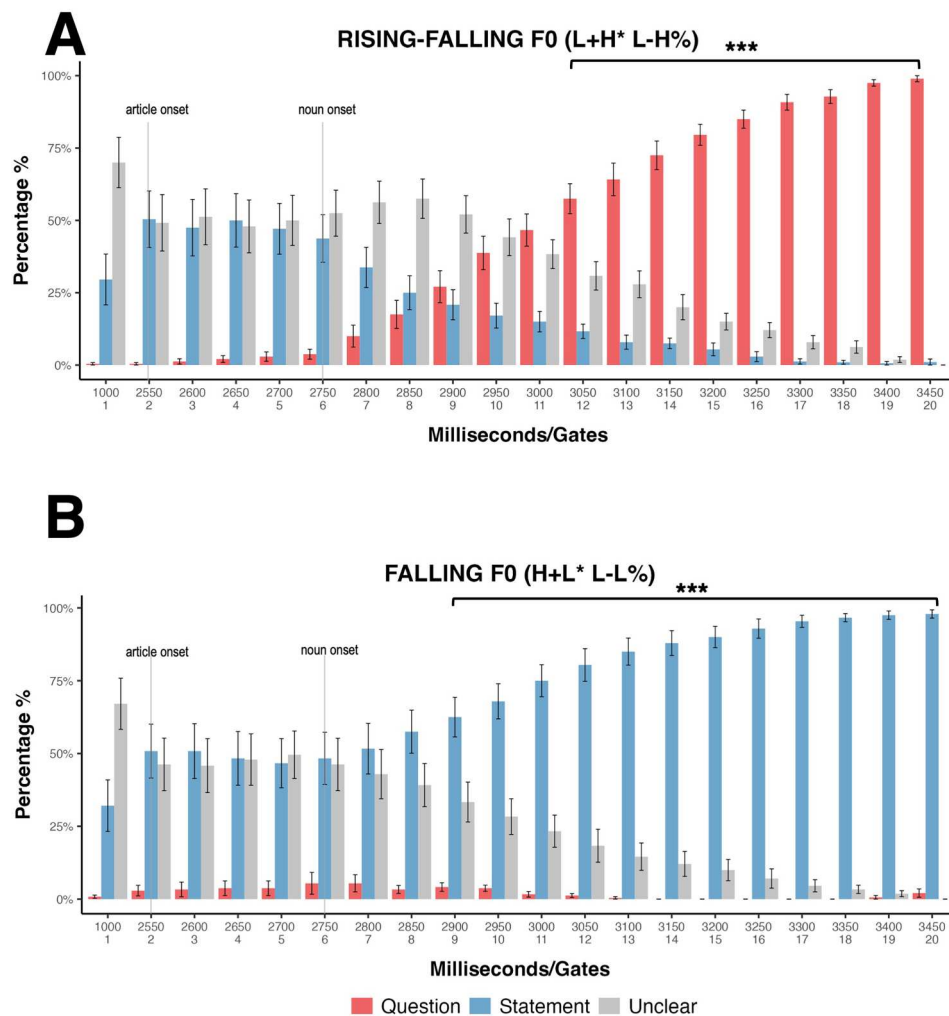


Figure 2. The time course of speech act recognition of sentences with a rising-falling F0 contour (A) and falling F0 contour (B) within the final word (noun onset) in cross-spliced utterances as a function of gate intervals (milliseconds). Specifically, the rising-falling F0 contour was characterised by a rising-falling pattern with an extra rise at the boundary tone (L + H* L-H%), while for the falling F0 contour refers to the falling pattern (H + L* L-L%; see Figure 1). The bar graphs depict the percentage of selected responses by participants for questions (first bar in red colour) and statements (second bar in blue) with confidence scores $\geq 70\%$ and, unclear along with all trials with confidence scores $< 70\%$ (third bar in grey). Error bars show standard error; asterisks indicate a significant difference between response types (Tukey multiple comparison tests).

this study was designed to (i) determine the minimum amount of speech and prosodic signals required for the recognition of communicative actions, which we refer to as the “speech act recognition point” within the gate construction; (ii) assess whether the rising-falling F0 patterns at the nuclear accents might play a role in this regard; and (iii) identify any (dis)similarities in the recognition points between speech acts in cross-spliced, well-matched stimuli and naturally spoken ones that exhibit variations in their entire F0 contours. We found that the recognition point was detected at the final word in both cross-spliced and natural

sentences, driven by the nuclear accentual rise/fall, with statements being recognised earlier compared to questions. Moreover, early response to incomplete sentence showed a bias towards interpreting them as statements, which was especially evident in cross-spliced stimuli, but also present in the natural sentences. This suggests that subjects were interpreting incomplete sentences as statements, a phenomenon we refer to as a “statement bias”. However, this bias appears very variable, as evident from visual observation of interindividual frequency response patterns. Additionally, differences in the time course of question recognition

Natural spoken sentences

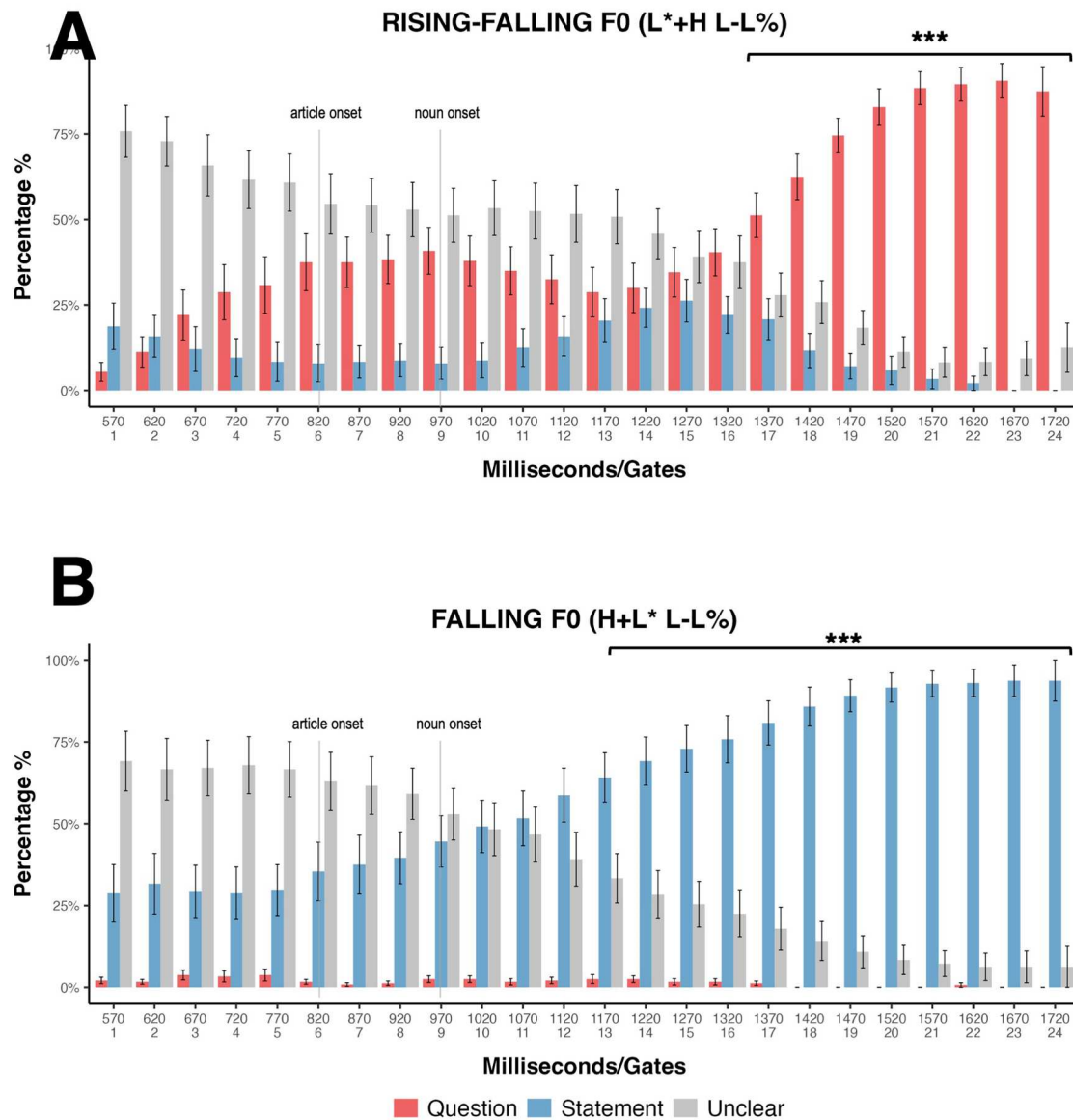


Figure 3. The time course of speech act recognition of sentences with a rising-falling F0 contour (A) and falling F0 contour (B) within the final word (noun onset) in natural utterances as a function of gate intervals (milliseconds). The rising-falling F0 contour was characterised by a rising-falling pattern with a rise after the stressed syllable and a low boundary tone (L*+H L-L%), while the falling F0 contour follows a falling pattern (H + L* L-L%). The bar graphs depict the percentage of selected responses by participants for questions (first bar in red colour) and statements (second bar in blue) with confidence scores $\geq 70\%$ and unclear along with all trials with confidence scores $< 70\%$ (third bar in grey). Error bars show standard error; asterisks indicate a significant difference between response types (Tukey multiple comparison tests).

were observed between cross-spliced and natural sentences due to prefinal F0 differences present in the natural conditions, while statements showed a similar recognition time course in both conditions. Below, we discuss each of these results in more detail in the context of previous studies.

5.1. Speech act recognition and the role of vocal F0 contour

Previous studies in the Italian language have documented a rising and falling F0 contour for polar questions and statements, respectively. In particular, a common

characteristic for questions across various Italian regional varieties is being characterised by a rise at the nuclear accent ($L + H^*$) or directly after ($L^* + H$), typically followed by a low boundary tone ($L\%$). However, in some cases, a high boundary tone ($H\%$) may also occur (see table 4 in Savino, 2012). In contrast, statements are typically characterised by a nuclear accentual fall followed by low boundary tone ($H + L^* L + L$). The stimuli of the present study show similar F0 contour at the final word sentence for question and statement in both cross-spliced and natural sentences with some variations in detail discussed above (see also Sec. 2.4 for more details). To examine speech act recognition as speech unfolds, we adapt the gating paradigm for speech act recognition as speech unfolds using very short gates of 50 ms, however, within the constraints of the gate construction of the present study. Specifically, the findings suggest that the recognition points for both questions and statements are driven by the nuclear accentual rise or fall in the sentences' final word, regardless of whether the sentences were cross-spliced or natural.

In cross-spliced sentences, the nuclear accentual F0 rise occurred, on averaged, across all stimuli approximately at ca. 100 ms after the noun onset, aligning with the first syllable (e.g. “**me**-tro”, $L + H^* L + L\%$, gate 8/9). In contrast, in the natural sentences, the F0 rise appeared directly after the nuclear accent, at ca. 300 ms after the final noun onset ($L^* + H L - L\%$, gate 15/16). Notably, at these time points, the frequency response for questions start to increase, by reaching a significant difference as compared to statement and unclear responses at 150/250 ms (gate 12) for the cross-spliced, and at 350/400 ms (gate 17/18) for the natural ones (see Figures 2 and 3). The differences in timing at which the rise occurred in question functions between cross-spliced and natural stimuli are related to the fact that in the cross-spliced stimuli the last noun was recorded separately and later combined with the first segment of the sentence, while the natural sentences were recorded at once. Consequently, this may have implications for whether studies on prosody and language processing are investigated with continuously spoken sentences or with cross-spliced ones. Furthermore, previous studies have highlighted the importance of a rising in the boundary tone ($H\%$) for question recognition (Avesani, 1990; Bertinetto & Loporcaro, 2005; Chapalaz, 1964; D'Imperio, 2002). While this feature was present in the cross-spliced stimuli but absent in the natural ones, our findings suggest that this was not the primary cue for recognition. Rather, the nuclear accentual F0 rise or right after, a pattern consistently

observed across Italian varieties (Savino, 2012), played a crucial role.

For statements, the recognition approximately coincided with the major F0 fall after the accentual syllable (e.g. “**me**-tro”). Specifically, in the cross-spliced conditions, a substantial F0 drop is observed between gates 8 and 9, with an averaged decrease of 36 Hz across all stimuli compared to the preceding gate 7. This pronounced F0 decline at the nuclear accent aligns with the recognition of statements (gate 9), as indicated by an increase in frequency classifications identifying the utterance as a statement. Note that an earlier 26 Hz F0 drop occurred between gates 5 and 6 right before the noun onset, which did not result in a significant change in frequency responses, when compared to questions and unclear responses. In the natural condition, a similar pattern was observed; however, a notable statistical difference for statements occurred when the averaged minimum value of the F0 contour was reached, specifically at gate 13, or substantially after the large drop (57 Hz F0 drop between gate 7 and gate 8, see Table 1). Although the F0 fall at the accentual syllable seems to be critical for recognising statements, the exact point of recognition, directly after the major F0 drop or later, cannot be conclusively determined with the current dataset and gate construction.

Furthermore, in the natural sentences, differences in F0 contour between questions and statements were also present earlier, specifically at 100 ms after the onset of the second verb, precisely at the second stressed syllable (*stanno pren-DEN-do*, “they are **taking**”, gate 3), in which 50 ms gates were constructed from the verb onset. However, these early differences did not lead the participants to a clear recognition, but instead only resulted in a slight increase in frequency response towards a question (more on this below). Hence, these findings suggest that intonational patterns, specifically at the nuclear accent at the sentence-final word, is a relevant marker driving the participants to clearly discriminate between the two speech acts. Note however, that the results are based not on the direct comparison between cross-spliced vs natural one but rather on between speech acts within each respective condition. As mentioned above, a direct comparison was not possible due to the differences introduced by the splicing technique, pause, duration, and F0 contour intonational patterns (see Sec. 2.4). Moreover, the interpretations are based on the average F0 modulations presented in Table 1, not taking into account variations within the stimuli or other specific acoustic features that could have played a role in recognition. However, the present study provides a first step in

investigating the time course of speech act recognition and a valuable approach for further research in this area, particularly through the use of the gating paradigm.

5.2. Time course of statement and question recognition and the statement bias

Despite both speech acts being identified after the onset of the last word, driven by the nuclear accentual F0 modulation, differences in recognition points appeared between the two speech acts: in the cross-spliced condition, statements were recognised about 150 ms (gate 9) earlier than questions (gate 12), while in the naturally spoken sentences, statements were recognised about 200 ms (gate 13) earlier than questions (gate 17, 350 ms). A possible explanation for the earlier detection of the statement can be found through an analysis of the frequency responses across gates. Intriguingly, the data show a preference towards statements in the first sentence segment, specifically from the second verb onset (“*stanno prendendo la ...*”, “*they are taking the ...*”) in both statements and questions in the cross-spliced stimuli (see the similar frequency response between the two in the first sentence segment in Figure 2). Thus, this suggests that participants initially tend to classify all sentence types in the cross-spliced condition mainly as statements, and only when the nuclear accentual F0 rise is encountered do they correctly identify sentences as having a question function. A possible plausible interpretation of this “statement bias” is that, in the absence of explicit pragmatic cues, here, prosodic ones, listeners may adopt a default interpretation of the utterance as a statement. Once sufficient prosodic information (e.g. a rising F0 contour) becomes available, they revise their understanding toward a non-declarative function, as a question. This seems to suggest that the statement may serve as the unmarked, default communicative function. Alternative explanations are also possible. The “statement bias” in the spliced condition may stem from the first two sentence segments being identical between statement and questions prior to the final word and possibly to their intonational pattern resembling declarative sentences (H + L* L + L %, i.e. falling contour), and/or for the lacking any F0 variations that are instead present in naturally spoken sentences. Although, these factors may also play a role, they are unlikely to fully account for it, as a bias is also observed in the natural sentences (see below). Furthermore, a closer visual examination of individual participants’ responses reveals notable variability in frequency patterns during the early segment of the sentence in the cross-spliced condition. Specifically, 6 out

of 16 participants demonstrate a clear statement bias, while the remaining participants exhibit a higher frequency of uncertainty in their responses (see Figures 7 and 8, Appendix 4 in supplementary material). Similarly, for statement sentences, variability is evident in the early segment, with 7 participants showing higher frequency responses for statements, potentially contributing to the observed early statement recognition in the averaged data across all subjects. Notably, the same subjects (#2,3,7,10,11,13) show a statement bias for both speech acts during the early segment, which could be additionally influenced by the regional linguistic differences among the participants tested (see Methods section), among the other factors mentioned above. Importantly, consistent frequency responses across all interindividual responses appear to be within the last word of the sentence, thus in line with the analysis on the averaged frequency responses.

For natural sentences, an initial “statement bias” is also observed for question sentences on the averaged response frequency across subjects, appearing as early as the first gate during the initial verb (“*Stanno* + first syllables of the second verb”) and persisting in early gates for statements (Figure 3(A)). However, the time course question recognition patterns showed a different trend as compared to cross spliced condition. Initially, participants identified the sentences as statements in gate 1 but began choosing “question” more frequently from gate 3 onward (see Figure 3(A)). This shift became slightly stronger after the first F0 rise at the second verb. However, between the noun onset and the nuclear accentual F0 rise, there was a decline in question responses and an increase in statement responses, suggesting a moment of uncertainty or hesitation in recognising the speech act. It is at the last word, when the nuclear accentual F0 contour is encountered, that a reliable recognition point with high certainty is identified, as indicated by the frequency averaged responses. Such different time course recognition in natural questions, might be due to the higher F0 rise for questions as compared to statements in this early segment. However, closer examination of individual responses also reveals variability in the early part of the sentence for the natural sentences. In particular, 3 participants (#9,11,12) show a clear preference for questions and others show an increase in question responses that remain comparable to the unclear responses (#6, 10, 16) and the rest an increase towards unclear responses. This variability may explain the higher frequency responses for questions prior the final word observed in the averaged responses, though it is not observed consistently in all subjects. Also, for statements, variability is present, with 6 participants (#2,3,7,9,10,12), showing higher frequency response for

statement, often comparable to the unclear responses. This may have contributed to the early recognition of statements in the averaged responses within the last word of the sentence, similar to what was observed for the cross-spliced condition (see figures 9 and 10 Appendix 4 in Supplementary materials). Overall, the pattern of speech act recognition as the sentence unfolds is variable between subjects in the early segment, again potentially influenced by participants originating from different Italian regions. Despite the early variability, both questions and statements are consistently and reliably recognised following the onset of the critical word, driven approximately by the rise around the nuclear accent. Finally, the loudness or intensity in the article “la” (the) prior to the final critical noun was stronger for statements compared to questions in both conditions. However, this difference didn’t affect the recognition, as no differences were found in the frequency response in those gates.

Based on these findings, the splicing technique, although effective for controlling acoustic differences, may have introduced artefacts, in particular, the statement bias present in cross-spliced questions or the differing recognition time course in natural questions (in both cases, however, present only in some of the subjects). Note, that both cross-spliced and natural conditions show similarities in the actual speech act recognition point, which seems to be driven by the nuclear accentual F0 modulation. This suggests that the splicing does not impact conscious recognition, but rather may affect the processes leading to it. However, more work is needed in this regard, specifically a more closed examination of interindividual responses that here were discussed only based on visual observation, due to the small sample size across participants from different Italian regions. In general, future neurolinguistic studies should consider the methodological implications of well-matching acoustic features through the splicing technique, which risks minimising some acoustic differences that could be relevant in communicative function processing. Specifically, the initial F0 prosodic modulation observed in natural sentences seems to serve as an indicator of the speech act type, thereby enabling some of the listeners to anticipate and prepare for rapid responses, as evidenced in investigations of turn-taking timing (e.g. Levinson, 2016; Meyer, 2023).

Furthermore, naturalness ratings – where participants rated how natural the stimuli sounded if encountered in real communicative interactions – showed no significant statistical interactions between the naturalness scores of cross-spliced and natural conditions. However, cross-spliced stimuli were generally perceived as less natural compared to the spoken sentence recorded as natural as possible. This expected result is likely due to the

application of the splicing technique to the stimuli and/or the cross-splicing of the article “la” between questions and statements and/or the longer pause prior the article + final word. Nevertheless, questions were rated on average as less natural than statements, regardless of whether the stimuli are cross-spliced or natural. Again, visual observation of inter-individual responses to naturalness ratings shows variability between subjects (see Supplementary Material Appendix 5, Figure 12 and Figure 13), so further work is needed in this regard as well. However, an explanation of the differences in naturalness between questions and statements might be due to their presentation in isolation in the present study, rather than embedded in their typical action schema in social interaction (more about these below, see also Tomasello 2023).

5.3. Relationship between gating and neurophysiological results

In our previous neurophysiological EEG study that used the same cross-spliced spoken sentences, a clear and significant brain response differences was found starting already 100 ms after the onset of the sentence-final noun phrase, coinciding with the nuclear accentual F0 rise (see Fig. 3A in Tomasello et al., 2022). This temporal distance was ca. 50 ms earlier than the predominance of statement over question judgements and even 150 ms earlier than the earliest question over statement judgement result obtained in the gating study. This difference may reflect a distinction between the earliest perceptual and classification processes in the brain’s language systems and the conscious recognition of the related differences and their implications for linguistic-pragmatic understanding. This said, we need to emphasise that the brain response could, in principle, reflect just the physical side of the difference between signals, independent of the change in communicative function central to this work. However, the ERP study also used control conditions with the same, or similar, physical change but without the difference in communication function understanding, where the neurophysiological difference was absent (Tomasello et al., 2022). This pattern of results argues in favour of an early ERP correlate of speech act understanding, which precedes the here documented difference in conscious speech act classification occurring later. The delay of recognition might be due to the fact that the gating task required participants to consciously decide on the type of speech act and their confidence scores. This decision-making process may add noise to the speech act recognition itself, affecting participants’ processing of the pragmatic meaning of sentence functions differently

from the passive task used in the EEG study. However, both neural and behavioural evidence suggest that comprehension and recognition are primarily driven by the nuclear accentual F0 modulation, rather than by later cues, such as the boundary tone. Crucially, this finding aligns with the speech act recognition results of the natural sentences in the present study.

5.4. Limitations and future perspectives

The results of the present study were interpreted as a consequence of F0 contour and its role in the recognition of different speech act types within the gating construction. However, other acoustic features might be present between the two conditions and speech acts that could have contributed to the results of the present gating study, such as other acoustic and phonetic cues on F0 contour or local and global speech rate (e.g. Bertinetto & Loporcaro, 2005; Cangemi & D'Imperio, 2011; Crocco, 2013; D'Imperio, 2002; Gili Fivela & Savino, 2003). Moreover, natural spoken sentences, which were left unaltered after recording, might differ more than cross-spliced sentences in those acoustic features. Thus, future studies should explore the contribution of additional factors in speech by employing a gating construction that starts from the onset of the sentence and aligned with specific intonational features, such as syllables, pitch accents, phrase accents, and boundary tones, rather than approximating it based on article and word onsets, as done in the present study. This refined approach may provide a more detailed understanding of how various intonational patterns and acoustic features interact to facilitate speech act recognition. Note however, that for the present study, we decide to have fixed 50 ms gates as to make it more comparable to the previous EEG study, which measures brain responses in constant time point intervals. Additionally, it would be valuable to specifically examine differences from participants coming from different Italian regions, which here was only possible to certain extent and only visually due to the limited number of participants from different Italian regions.

Furthermore, the present study examined speech acts in isolation not including the typical actions following a question or a statement functions. Specifically, questions are intrinsically tied to the expectation of the partner answering the questions, while statements typically trigger a sign of confirmation of receipt (e.g. by nodding, among others). These action sequence structure have been claimed to be a critical part of a mental representation of speech acts and possibly play a critical role in the understanding processes (Pulvermüller et al., 2014; Tomasello, 2023). Thus, future studies using the same

method should embed speech acts in their typical action schema during social interaction and examine if recognition and discrimination might lead to different results from the present one. Specifically, researchers should investigate whether the naturalness rating, which currently shows questions being perceived as less natural than statements, would differ if speech acts are presented within their typical interactional context.

Finally, future studies could adapt the present experimental design to investigate possible interactions between prosody and other markers of speech act function on comprehension, and thus on the recognition point of speech acts. For instance, multimodal cue processing could be examined, such as the influence of different facial expressions, and/or gestures (Crespo-Sendra et al., 2013; Holler & Levinson, 2019; Nota et al., 2022; Özyürek, 2021; Tomasello et al., 2019), to determine whether these elements might speed up recognition and decision-making during the task. In addition, it would be worthwhile to explore the role of other contextual information, such as background assumptions or common ground and the difference between direct and indirect speech acts recognition (e.g. Boux et al., 2023; see for a review Tomasello et al., 2025) and other linguistic actions we can perform with words and sentences (Searle, 1979). Sentences could be embedded in dialogues or short stories with the aim of providing some contextual knowledge and assessing its effect on speech act recognition. For instance, if the context makes it clear that the speaker knows that p, then it is unlikely that the speaker's next speech act is asking whether p (question). Notably, context and ambiguity can make intonational cues less necessary, especially in languages where syntactic structure provides a clear marker for speech act recognition. Investigating the interactions between speech act marking by prosody and by other linguistic pragmatic factors is a necessary step towards a better understanding of speech act recognition in naturalistic, social interactions. Overall, further investigations with stimuli perhaps embedded in more natural contexts and additional larger sample sizes are needed that may unfold further insights on the influence of vocal intonation processing over the exact recognition point of speech acts.

6. Conclusion

The present study explores how speech prosody is used to convey communicative function and how they are recognised by the listeners during the temporal unfolding of the same Italian sentences by applying the gating paradigm. Our data show that the difference between statements and yes/no questions, indicated by falling

and rising-falling F0 contours respectively, is accurately detected 150–350 ms after the onset of the sentence-final word, corresponding around the nuclear accentual F0 modulation at the first syllables. The data also show that statements are recognised slightly earlier than questions and discuss the possible reasons for that, including visual inspection of interindividual responses. We also highlight potential artefacts that can be introduced by splicing techniques, as some gradual acoustic and prosodic variability is present in natural speech long before the main prosodic difference carrying communicative function information kicks in. Overall, the gating paradigm has proven to be an adequate tool for studying how recognition of speech acts conveyed by speech prosody evolves over time and provides initial insights on the time course of communicative action understanding as speech unfolds.

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Data availability statement

All data and scripts are available at: <https://osf.io/c83ep/>.

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References

- Agard, F. B., & Di Pietro, R. J. (1965). *The sounds of English and Italian*. University of Chicago Press.
- Antoine, S., Grisoni, L., Tomasello, R., & Pulvermüller, F. (2024). The prediction potential indexes the meaning and communicative function of upcoming utterances. *Cortex*, 177, 346–362. <https://doi.org/10.1016/j.cortex.2024.05.011>
- Arvaniti, A. (2022). The autosegmental-metrical model of intonational phonology. In S. Shattuck-Hufnagel & J. Barnes (Eds.), *Prosodic theory and practice* (pp. 25–84). MIT Press. <https://doi.org/10.7551/mitpress/10413.003.0004>
- Astésano, C., Besson, M., & Alter, K. (2004). Brain potentials during semantic and prosodic processing in French. *Cognitive Brain Research*, 18(2), 172–184. <https://doi.org/10.1016/j.cogbrainres.2003.10.002>
- Audibert, N., Auberge, V., & Rilliard, A. (2007, August 6–10). When is the emotional information? A gating experiment for gradient and contours cues. In *Proceedings of ICPHS XVI Meeting* (pp. 2137–2140).
- Austin, J. L. (1975). *How to do things with words*. Oxford University Press.
- Avesani, C. (1990). A contribution to the synthesis of Italian intonation. In *Proceedings of ICSLP 90, International Conference on Spoken Language Processing* (pp. 833–836).
- Avesani, C. (1995). ToBlt: Un sistema di trascrizione per l’intonazione italiana. In *Atti delle 5e Giornate di Studio del Gruppo di Fonetica Sperimentale (A.I.A.)* (pp. 85–98).
- Banse, R., & Scherer, K. R. (1996). Acoustic profiles in vocal emotion expression. *Journal of Personality and Social Psychology*, 70(3), 614–636. <https://doi.org/10.1037/0022-3514.70.3.614>
- Bänziger, T., & Scherer, K. R. (2005). The role of intonation in emotional expressions. *Speech Communication*, 46(3–4), 252–267. <https://doi.org/10.1016/j.specom.2005.02.016>
- Bertinetto, P. M., & Loporcaro, M. (2005). The sound pattern of Standard Italian, as compared with the varieties spoken in Florence, Milan and Rome. *Journal of the International Phonetic Association*, 35(2), 131–151. <https://doi.org/10.1017/S0025100305002148>
- Boersma, P. (2001). Praat, a system for doing phonetics by computer. *Glott International*, 5, 341–345.
- Bolinger, D. (1978). Intonation across languages. In J. H. Greenberg (Ed.), *Universals of human languages* (pp. 471–524). Stanford University Press.

- Bolinger, D. (1986). *Intonation and its parts: Melody in spoken English*. Stanford University Press.
- Boux, I. P., Margiotoudi, K., Dreyer, F. R., Tomasello, R., & Pulvermüller, F. (2023). Cognitive features of indirect speech acts. *Language, Cognition and Neuroscience*, 38(1), 40–64. <https://doi.org/10.1080/23273798.2022.2077396>
- Boux, I., Tomasello, R., Grisoni, L., & Pulvermüller, F. (2021). Brain signatures predict communicative function of speech production in interaction. *Cortex*, 135, 127–145.
- Cangemi, F., & D'Imperio, M. (2011). Local speech rate differences between questions and statements in Italian. In *ICPhS* (pp. 392–395).
- Chapallaz, M. (1964). Notes on the intonation of questions in Italian. In D. Abercrombie, D. B. Fry, P. A. D. MacCarthy, N. C. Scott, & J. L. M. Trim (Eds.), *Honour of Daniel Jones: Papers contributed in the occasion of his eightieth birthday* (pp. 306–312). Longmans.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates, Publishers.
- Cornew, L., Carver, L., & Love, T. (2010). There's more to emotion than meets the eye: A processing bias for neutral content in the domain of emotional prosody. *Cognition and Emotion*, 24(7), 1133–1152. <https://doi.org/10.1080/02699930903247492>
- Crespo-Sendra, V., Kaland, C., Swerts, M., & Prieto, P. (2013). Perceiving incredulity: The role of intonation and facial gestures. *Journal of Pragmatics*, 47(1), 1–13. <https://doi.org/10.1016/j.pragma.2012.08.008>
- Crocco, C. (2013). Is Italian clitic right dislocation grammaticalised? A prosodic analysis of yes/no questions and statements. *Lingua. International Review of General Linguistics. Revue Internationale De Linguistique Generale*, 133, 30–52. <https://doi.org/10.1016/j.lingua.2013.03.003>
- Cruttenden, A. (1997). *Intonation*. Cambridge University Press.
- D'Andrea, A., Ferri, F., & Grifoni, P. (2014). Prosodic analysis: An Italian case study. *International Journal of Language Studies*, 8(4).
- D'Imperio, M. (2002). Italian intonation: An overview and some questions. *Probus (Special Volume on Intonation in Romance Languages)*, 14(1), 37–69.
- Egorova, N., Pulvermüller, F., & Shtyrov, Y. (2014). Neural dynamics of speech act comprehension: An MEG study of naming and requesting. *Brain Topography*, 27(3), 375–392. <https://doi.org/10.1007/s10548-013-0329-3>
- Egorova, N., Shtyrov, Y., & Pulvermüller, F. (2013). Early and parallel processing of pragmatic and semantic information in speech acts: Neurophysiological evidence. *Frontiers in Human Neuroscience*, 7, 86. <https://doi.org/10.3389/fnhum.2013.00086>
- Escandell-Vidal, V. (2012). Prosodia y pragmática. *Studies in Hispanic and Lusophone Linguistics*, 4(1), 1–14.
- Escandell-Vidal, V., & Prieto, P. (2020). Pragmatics and prosody in research on Spanish. In D. A. Koike, & J. C. Félix-Brasdefer (Eds.), *The Routledge handbook of Spanish pragmatics* (pp. 149–166). Routledge.
- Falé, I., & Faria, I. H. (2006). Time course of intonation processing in European Portuguese: A gating paradigm approach. In *IV Jornadas en tecnologia del Habla* (pp. 265–270).
- Fiorelli, P. (1965). Corso di pronunzia italiana. *Ed Ràdar*. 190.
- Frick, R. W. (1985). Communicating emotion: The role of prosodic features. *Psychological Bulletin*, 97(3), 412–429. <https://doi.org/10.1037/0033-2909.97.3.412>
- Frota, S. (2002). Nuclear falls and rises in European Portuguese: A phonological analysis of declarative and question intonation. *Probus*, 14(1), 113–146. <https://doi.org/10.1515/prbs.2002.001>
- Gili Fivela, B., & Savino, M. (2003). Segments, syllables and tonal alignment: A study on two varieties of Italian. In *Proceedings of the 15th International Conference on Phonetic Sciences (ICPhS 2003)* (pp. 2933–2936).
- Grice, M., D'Imperio, M., Savino, M., & Avesani, C. (2005). Strategies for intonation labelling across varieties of Italian. *Prosodic Typology: The Phonology of Intonation and Phrasing*, 1, 362–389. <https://doi.org/10.1093/acprof:oso/9780199249633.003.0013>
- Grosjean, F. (1980). Spoken word recognition processes and the gating paradigm. *Perception & Psychophysics*, 28(4), 267–283. <https://doi.org/10.3758/BF03204386>
- Grosjean, F. (1985). The recognition of words after their acoustic offset: Evidence and implications. *Perception & Psychophysics*, 38(4), 299–310. <https://doi.org/10.3758/BF03207159>
- Grosjean, F. (1996). Gating. *Language & Cognitive Processes*, 11(6), 597–604. <https://doi.org/10.1080/016909696386999>
- Hellbernd, N., & Sammler, D. (2016). Prosody conveys speaker's intentions: Acoustic cues for speech act perception. *Journal of Memory and Language*, 88, 70–86. <https://doi.org/10.1016/j.jml.2016.01.001>
- Hellbernd, N., & Sammler, D. (2018). Neural bases of social communicative intentions in speech. *Social Cognitive and Affective Neuroscience*, 13(6), 604–615. <https://doi.org/10.1093/scan/nsy034>
- Holler, J., & Levinson, S. C. (2019). Multimodal language processing in human communication. *Trends in Cognitive Sciences*, 23(8), 639–652. <https://doi.org/10.1016/j.tics.2019.05.006>
- Holtgraves, T. (2008). Automatic intention recognition in conversational processing. *Journal of Memory and Language*, 58(3), 627–645. <https://doi.org/10.1016/j.jml.2007.06.001>
- Holtgraves, T. (2021). Understanding miscommunication: Speech act recognition in digital contexts. *Cognitive Science*, 45(10), e13023. <https://doi.org/10.1111/cogs.13023>
- Howes, D. (1957). On the relation between the intelligibility and frequency of occurrence of English words. *The Journal of the Acoustical Society of America*, 29(2), 296–305. <https://doi.org/10.1121/1.1908862>
- Jiang, X., & Pell, M. D. (2015). On how the brain decodes vocal cues about speaker confidence. *Cortex*, 66, 9–34. <https://doi.org/10.1016/j.cortex.2015.02.002>
- Kotz, S. A., & Paulmann, S. (2007). When emotional prosody and semantics dance cheek to cheek: ERP evidence. *Brain Research*, 1151, 107–118. <https://doi.org/10.1016/j.brainres.2007.03.015>
- Ladd, D. R. (2008). *Intonational phonology*. Cambridge University Press.
- Levinson, S. C. (2016). Turn-taking in human communication—origins and implications for language processing. *Trends in Cognitive Sciences*, 20(1), 6–14. <https://doi.org/10.1016/j.tics.2015.10.010>
- Lieberman, P. (1967). *Intonation, perception, and language*. MIT Research Monograph.
- Maiden, M., & Robustelli, C. (2014). *A reference grammar of modern Italian*. Routledge.
- Marslen-Wilson, W. D. (1984). Function and process in spoken word recognition: A tutorial review. In H. Bouma & D. Bouwhuis (Eds.), *Attention & performance* (Chap. X, pp. 125–150). Lawrence Erlbaum Associates.

- Marslen-Wilson, W. D. (1987). Functional parallelism in spoken word-recognition. *Cognition*, 25(1–2), 71–102. [https://doi.org/10.1016/0010-0277\(87\)90005-9](https://doi.org/10.1016/0010-0277(87)90005-9)
- Marslen-Wilson, W. D., & Welsh, A. (1978). Processing interactions during word-recognition in continuous speech. *Cognitive Psychology*, 10(1), 29–63. [https://doi.org/10.1016/0010-0285\(78\)90018-X](https://doi.org/10.1016/0010-0285(78)90018-X)
- Mauchand, M., & Pell, M. D. (2022). Listen to my feelings! How prosody and accent drive the empathic relevance of complaining speech. *Neuropsychologia*, 175, Article 108356. <https://doi.org/10.1016/j.neuropsychologia.2022.108356>
- Metsala, J. L. (1997). An examination of word frequency and neighborhood density in the development of spoken-word recognition. *Memory & Cognition*, 25(1), 47–56. <https://doi.org/10.3758/BF03197284>
- Meyer, A. S. (2023). Timing in conversation. *Journal of Cognition*, 6(1), 20. <https://doi.org/10.5334/joc.268>
- Monetta, L., Cheang, H. S., & Pell, M. D. (2008). Understanding speaker attitudes from prosody by adults with Parkinson's disease. *Journal of Neuropsychology*, 2(2), 415–430. <https://doi.org/10.1348/174866407X216675>
- Nota, N., Trujillo, J. P., & Holler, J. (2022). Conversational eyebrow frowns facilitate question identification: An online VR study. *PsyArXiv*. <https://doi.org/10.31234/osf.io/fcj8b>
- O'Shaughnessy, D. (1979). Linguistic features in fundamental frequency patterns. *Journal of Phonetics*, 7(2), 119–145. [https://doi.org/10.1016/S0095-4470\(19\)31045-9](https://doi.org/10.1016/S0095-4470(19)31045-9)
- Özyürek, A. (2021). Considering the nature of multimodal language from a crosslinguistic perspective. *Journal of Cognition*, 4(1), 42. <https://doi.org/10.5334/joc.165>
- Pell, M. D., Jaywant, A., Monetta, L., & Kotz, S. A. (2011). Emotional speech processing: Disentangling the effects of prosody and semantic cues. *Cognition & Emotion*, 25(5), 834–853. <https://doi.org/10.1080/02699931.2010.516915>
- Pell, M. D., & Kotz, S. A. (2011). On the time course of vocal emotion recognition. *PLoS One*, 6(11), e27256. <https://doi.org/10.1371/journal.pone.0027256>
- Petrone, C., & Niebuhr, O. (2014). On the intonation of German intonation questions: The role of the prenuclear region. *Language and Speech*, 57(1), 108–146. <https://doi.org/10.1177/0023830913495651>
- Pierrehumbert, J. (1981). Synthesizing intonation. *The Journal of the Acoustical Society of America*, 70(4), 985–995. <https://doi.org/10.1121/1.387033>
- Pulvermüller, F., Moseley, R. L., Egorova, N., Shebani, Z., & Boulenger, V. (2014). Motor cognition–motor semantics: Action perception theory of cognition and communication. *Neuropsychologia*, 55, 71–84. <https://doi.org/10.1016/j.neuropsychologia.2013.12.002>
- R Core Team. (2019). *R: A language and environment for statistical computing (version 3.5.2, R Foundation for Statistical Computing, Vienna, Austria, 2018)*. R Foundation for Statistical Computing.
- Rigoulot, S., Fish, K., & Pell, M. D. (2014). Neural correlates of inferring speaker sincerity from white lies: An event-related potential source localization study. *Brain Research*, 1565, 48–62. <https://doi.org/10.1016/j.brainres.2014.04.022>
- Rosenzweig, M. R., & Postman, L. (1957). Intelligibility as a function of frequency of usage. *Journal of Experimental Psychology*, 54(6), 412. <https://doi.org/10.1037/h0041465>
- Rubenstein, H., & Pollack, I. (1963). Word predictability and intelligibility. *Journal of Verbal Learning and Verbal Behavior*, 2(2), 147–158. [https://doi.org/10.1016/S0022-5371\(63\)80079-1](https://doi.org/10.1016/S0022-5371(63)80079-1)
- Salasoo, A., & Pisoni, D. B. (1985). Interaction of knowledge sources in spoken word identification. *Journal of Memory and Language*, 24(2), 210–231. [https://doi.org/10.1016/0749-596X\(85\)90025-7](https://doi.org/10.1016/0749-596X(85)90025-7). PMID: 23226691; PMCID: PMC3513696.
- Sammler, D., Grosbras, M. H., Anwender, A., Bestelmeyer, P. E., & Belin, P. (2015). Dorsal and ventral pathways for prosody. *Current Biology*, 25(23), 3079–3085. <https://doi.org/10.1016/j.cub.2015.10.009>
- Savino, M. (2012). The intonation of polar questions in Italian: Where is the rise? *Journal of the International Phonetic Association*, 42(1), 23–48. <https://doi.org/10.1017/S002510031100048X>
- Schaerlaeken, S., & Grandjean, D. (2018). Unfolding and dynamics of affect bursts decoding in humans. *PLoS One*, 13(10), e0206216. <https://doi.org/10.1371/journal.pone.0206216>
- Searle, J. R. (1979). *Expression and meaning: Studies in the theory of speech acts*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511609213>
- Simon-Thomas, E. R., Keltner, D. J., Sauter, D., Sinicropi-Yao, L., & Abramson, A. (2009). The voice conveys specific emotions: Evidence from vocal burst displays. *Emotion*, 9(6), 838–846. <https://doi.org/10.1037/a0017810>
- Srinivasan, R. J., & Massaro, D. W. (2003). Perceiving prosody from the face and voice: Distinguishing statements from echoic questions in English. *Language and Speech*, 46(1), 1–22. <https://doi.org/10.1177/00238309030460010201>
- Steinberg, J., Truckenbrodt, H., & Jacobsen, T. (2012). The role of stimulus cross-splicing in an event-related potentials study. Misleading formant transitions hinder automatic phonological processing. *The Journal of the Acoustical Society of America*, 131(4), 3120–3140. <https://doi.org/10.1121/1.3688515>
- Steinhauer, K., Alter, K., & Friederici, A. D. (1999). Brain potentials indicate immediate use of prosodic cues in natural speech processing. *Nature Neuroscience*, 2(2), 191–196. <https://doi.org/10.1038/5757>
- Studdert-Kennedy, M., & Hadding, K. (1973). Auditory and linguistic processes in the perception of intonation. *Language and Speech*, 16(4), 293–313. <https://doi.org/10.1177/002383097301600401>
- Tomasello, R. (2023). Linguistic signs in action: The neuropragmatics of speech acts. *Brain and Language*, 236, Article 105203. <https://doi.org/10.1016/j.bandl.2022.105203>
- Tomasello, R., Boux, I. P., & Pulvermüller, F. (2025). Theory of mind and the brain substrates of direct and indirect communicative action understanding. *Philosophical Transactions of the Royal Society B*. <https://doi.org/10.1098/rstb.2023.0497>
- Tomasello, R., Grisoni, L., Boux, I., Sammler, D., & Pulvermüller, F. (2022). Instantaneous neural processing of communicative functions conveyed by speech prosody. *Cerebral Cortex*, 32(21), 4885–4901. <https://doi.org/10.1093/cercor/bhab522>
- Tomasello, R., Kim, C., Dreyer, F. R., Grisoni, L., & Pulvermüller, F. (2019). Neurophysiological evidence for rapid processing of verbal and gestural information in understanding communicative actions. *Scientific Reports*, 9(1), 1–17. <https://doi.org/10.1038/s41598-018-37186-2>
- Tukey, J. W. (1977). *Exploratory data analysis* (Vol. 2, pp. 131–160). Addison-Wesley.

Vieillard, S., Peretz, I., Gosselin, N., & Khalfa, S. (2008). Happy, sad, scary and peaceful musical excerpts for research on emotions. *Cognition and Emotion*, 22(4), 720–752. <https://doi.org/10.1080/02699930701503567>

Walley, A. C., Michela, V. L., & Wood, D. R. (1995). The gating paradigm: Effects of presentation format on spoken word recognition by children and adults. *Perception & Psychophysics*, 57(3), 343–351. <https://doi.org/10.3758/BF03213059>